

CRDTs and Databases: A Hands-On Tutorial

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Why this tutorial

- ▶ Isolation models with a total commit order, such as serializability or snapshot isolation, are not compatible with large-scale, highly available and partition-tolerant services.
- ▶ Even a weaker model such as parallel snapshot isolation, when defined over a read-write API, must either abort concurrent transactions on shared objects or lose updates.
- ▶ The way out is to offer transactions over **higher-level abstractions** beyond a read-write API. Conflict-free Replicated Data Types are such an abstraction.
- ▶ CRDTs and relational databases, however, have been developed in two worlds apart. Existing CRDT databases have ad hoc, non-relational implementations that do not combine well with query optimization.

What we cover

What CRDTs are, how CRDTs can be combined with the relational model, accessed by standard SQL, and implemented over standard relational databases.

Roadmap

(1) Introduction to CRDTs

60 min

- historical tour, consistency and the CAP theorem
- operation-based CRDTs
- state-based CRDTs
- comparison of approaches

(2) CRDTs in database systems

30 min

- key challenges and requirements
- a survey of existing implementations in databases
- Conflict-free Replicated Data Views (CRDVs)

(3) Hands-on with CRDVs

60 min

- a tour of CRDVs in PostgreSQL
- the CRDVs cookbook for application developers
- performance analysis

(4) Open challenges

15 min

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